



Spring 2021–2022 (Math 102: Calculus II)

Final Exam: Suggested solutions

1. **(10 Points)** The limit is indeterminate of the form $0 \cdot \infty$. In turn, the limit can be written as follows:

$\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{\cos(2x)}$, which has the form $\frac{0}{0}$. Applying l'Hospital's Rule, the latter gives:

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{-\sec^2 x}{-2 \sin(2x)} = \frac{(\sqrt{2})^2}{2} = 1.$$

2. **(10 Points)** Integrating by parts with $u = \tan^{-1}(4x)$ and $dv = dx$, we obtain:

$$\begin{aligned} \int \tan^{-1}(4x) \, dx &= x \tan^{-1}(4x) - \int x \frac{4}{1 + (4x)^2} \, dx = x \tan^{-1}(4x) - 4 \int \frac{x}{1 + 16x^2} \, dx \\ &= x \tan^{-1}(4x) - \frac{4}{32} \int \frac{1}{u} \, dx \quad (u = 1 + 16x^2) = x \tan^{-1}(4x) - \frac{1}{8} \ln(1 + 16x^2) + C, \end{aligned}$$

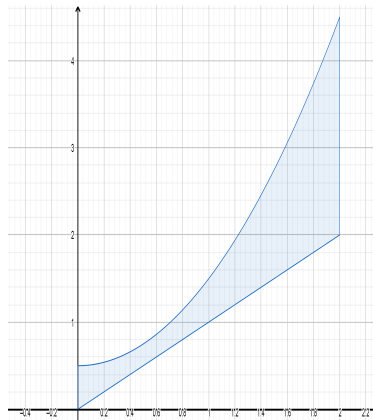
where C is a real constant.

3. **(10 Points)** This is a Geometric Series whose first term is $\sin x$ and common ratio $r = -\frac{\sin x}{2}$ and hence $|r| = \frac{1}{2} |\sin x| < 1/2$, for any real number x . Therefore, the series converges for all x and its sum is

$$s = \frac{\sin x}{1 + \frac{\sin x}{2}} = \frac{2 \sin x}{2 + \sin x}.$$

4. **(10 Points)** Using the disks method, we have:

$$V = \int_0^2 \pi \left(\left[\frac{1}{2} + x^2 \right]^2 - x^2 \right) dx = \pi \int_0^2 \left(x^4 + \frac{1}{4} \right) dx = \pi \left[\frac{x^5}{5} + \frac{x}{4} \right]_0^2 = 69\pi/10.$$

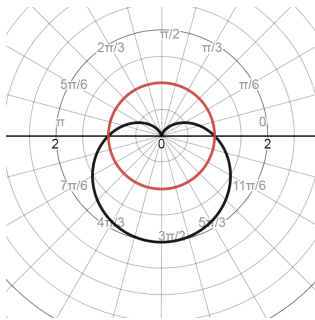


5. **(15 Points)** The two curves intersect when $1 - \sin \theta = 1$, which gives $\theta = k\pi$, $k = 0, 1, 2$, for $0 \leq \theta \leq 2\pi$.

Using the symmetry of the region about $\theta = \pi/2$, the area is

$$\begin{aligned} A &= 2 \int_{\pi}^{3\pi/2} \frac{1}{2} [(1 - \sin \theta)^2 - (1)^2] d\theta = \int_{\pi}^{3\pi/2} (\sin^2 \theta - 2 \sin \theta) d\theta \\ &= \int_{\pi}^{3\pi/2} \left(\frac{1 - \cos(2\theta)}{2} - 2 \sin \theta \right) d\theta = \left[\frac{\theta}{2} - \frac{\sin(2\theta)}{4} + 2 \cos \theta \right]_{\pi}^{3\pi/2} = 2 + \frac{\pi}{4}. \end{aligned}$$

NB: one can also integrate over the interval $[-\pi/2, 0]$.



6. **(10+10=20 Points)** (a) The exact length of the curve is

$$\begin{aligned} L &= \int_0^{\pi/4} \sqrt{1 + \left(\frac{dx}{dy} \right)^2} dy = \int_0^{\pi/4} \sqrt{1 + \left(\frac{\sec x \tan x}{\sec x} \right)^2} dx = \int_0^{\pi/4} \sqrt{1 + \tan^2 x} dx = \int_0^{\pi/4} \sec x dx \\ &= \ln |\sec x + \tan x| \Big|_0^{\pi/4} = \ln(\sqrt{2} + 1). \end{aligned}$$

- (b) The area of the surface is:

$$\begin{aligned} \int 2\pi y ds &= 2\pi \int_1^2 y \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dy = 2\pi \int_1^2 y \sqrt{1 + 16y^2} dy = 2\pi \int_{17}^{65} \frac{1}{32} \sqrt{u} du, \quad (u = 1 + 16y^2) \\ &= \frac{\pi}{16} \left[\frac{2}{3} u^{3/2} \right]_{17}^{65} = \frac{\pi}{24} (65\sqrt{65} - 17\sqrt{17}). \end{aligned}$$

7. **(10+5+10=25 Points)** (a) (i) The area of the surface is

$$\begin{aligned} S &= \int_0^1 2\pi y \sqrt{(dx/dt)^2 + (dy/dt)^2} dt = 2\pi \int_0^1 3t^2 \sqrt{(3-3t^2)^2 + (6t)^2} dt = 2\pi \int_0^1 3t^2 \sqrt{[3(t^2+1)]^2} dt \\ &= 18\pi \int_0^1 (t^4 + t^2) dt = 18\pi \left[\frac{t^5}{5} + \frac{t^3}{3} \right]_0^1 = 48\pi/5. \end{aligned}$$

- (ii) C has a vertical tangent line when $\frac{dx}{dt} = 3 - 3t^2 = 0$, that is, $t = \pm 1$ (note that $dy/dt \neq 0$ there).

Since $0 \leq t \leq 1$, then $t = 1$ is the only accepted value and the corresponding point on C is $(2, 3)$.

- (b) The curve is concave upward if $d^2y/dx^2 > 0$. First, we have:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-3 \sin t}{2 \cos t} = -\frac{3}{2} \tan t.$$

Then, it follows that

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{d}{dt} \left(-\frac{3}{2} \tan t \right)}{dx/dt} = \frac{-\frac{3}{2} \sec^2 t}{2 \cos t} = -\frac{3}{4} \sec^3 t.$$

Therefore, the concavity of the given curve is upward if $\sec t < 0$, for $0 < t < 2\pi$, that is, $\pi/2 < t < 3\pi/2$.